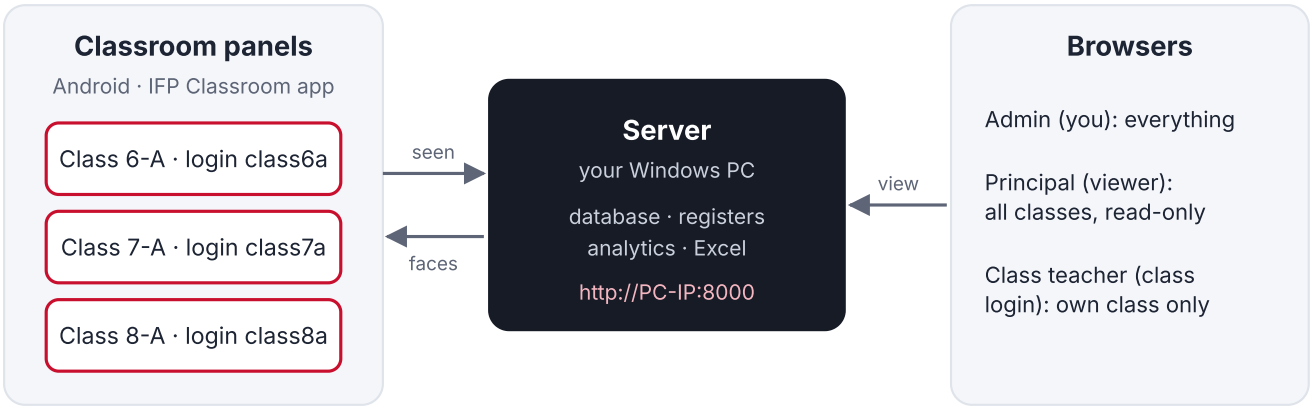
SETUP GUIDE & USER MANUAL

IFP Classroom Attendance

Face-recognition attendance and classroom entry / exit records using the camera on every classroom interactive flat panel



Server: Windows PC (school IT administrator) · **Clients:** IFP Classroom Android app on every classroom panel · **Viewing:** any browser on the school network

Software version: server with panel app 1.3. Screenshots use a fictional school ("Green Valley Public School") and sample names. Faces in camera images are blurred.

Face recognition uses the open-source face-api library (MIT licence), included with the software, so recognition works without internet access.

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Quick start (for the IT administrator)

1. On your Windows PC, install Python, then run `windows\install_windows_server.bat` as administrator (chapter 3).
2. Open `http://<PC-IP>:8000`, sign in as **admin / admin** and change the password (chapter 4).
3. Add every classroom and give each its **own login** (chapter 5). Add periods, holidays and weekly offs (chapter 6).
4. Import students and teachers from a CSV and add face photos (chapters 7–8).
5. On every panel, open `http://<PC-IP>:8000/app`, install the app and sign in with that classroom's login (chapter 9).
6. Check the registers and analytics from your PC (chapters 11–14). Give class teachers their classroom login and the principal a viewer login (chapter 15).

1. How the system works

IFP Classroom Attendance uses the camera on each classroom's interactive flat panel (IFP) to recognise students and teachers by face. It marks attendance and records when each person **enters** and **leaves** the classroom. Everything is stored on one **server**, your Windows PC, where all reports and analysis are available.

- **Server: your Windows PC.** It holds the database: classrooms, people, face data and entry/exit records. It produces the registers, the analytics and the Excel/CSV downloads.
- **Clients: the IFP Classroom Android app** on every classroom panel. The app recognises faces *on the panel* and sends only "person X was seen at time T". No video is sent or recorded.
- **One login per classroom, same server.** Each classroom has its own username and password, e.g. `class8a`. The panel app signs in with it, and the class teacher can use it in a browser to see *only that class*.
- **Viewing:** any PC, laptop or phone on the school network can open the server in a browser.

Key terms

Term	Meaning
Sighting	One "person seen in this classroom at this time" record sent by a panel. A person in view is recorded about every 30 seconds.
Entry	The first time a person is seen in a classroom, or the first sighting after they were away.
Exit	The last time a person was seen, once they have been out of view for the classroom's *exit after unseen* time (default 10 minutes). If they come back later, a new entry starts.
In class now	Seen recently, within the exit time, so no exit has been recorded yet.
Present / Absent	**Class register:** present if seen in that classroom that day. **Monthly register and Analytics:** present if seen in *any* classroom that day. Absent if not seen on a school day.
H / WO	Holiday / weekly off. Nobody is marked absent on these days.
Classroom login	The username and password of one classroom. Used by its panel app and its class teacher.
Kiosk link	A secret web address for a classroom's camera screen. The app gets it automatically when it signs in; you rarely need it yourself.

> **Attendance is always worked out from the raw sightings.**
> Changing a holiday, a weekly off, a period, a classroom's exit time or a person's class updates every report straight away. Nothing needs to be "recalculated".

2. What you need

Item	Requirement
Server PC	Windows 10 or 11, 4 GB RAM or more, about 2 GB free disk. It must stay switched on during school hours; set Sleep to *Never* when plugged in. Give it a **fixed IP address** (DHCP reservation in the router, or a static IP).
Software on the server	Python 3.10 or newer from python.org, installed with **Add Python to PATH** ticked. The installer downloads the rest the first time, so internet is needed once.
Classroom panels	Android 7 or newer IFPs with a built-in or USB camera. No OPS PC module is needed.
Network	Panels and the server PC on the same school network (LAN or Wi-Fi), with TCP port **8000** reachable on the server. The installer opens it in Windows Firewall.
Photos	One or more clear, front-facing photos of each student and teacher. ID-card photos are fine.
Consent	Parent/guardian consent for students' biometric data (see chapter 18).

What is in the `ifp-classroom` folder

Folder / file	Purpose
`windows\`	Installer, start-up script, daily backup and uninstaller for the Windows server.
`classroom_app\`	The server program, including the panel app file `static\app\ifp-classroom.apk`.
`data\` (created by the installer)	**All your data:** database, backups and logs. Back this folder up.
`android\`	Source of the panel app, only needed to rebuild it.
`docs\`	This guide and a text version (`SETUP.md`).

3. Install the server on your Windows PC

1. Install **Python 3.10 or newer** from `python.org/downloads`. On the first installer screen, tick **Add Python to PATH**.
2. Copy the `ifp-classroom` folder to the PC, e.g. `C:\IFPClassroom`. Keep it in a fixed place; the start-up task points to it.
3. Open the `windows` folder, right-click **install_windows_server.bat** and choose **Run as administrator**.
4. Wait for the four steps to finish. The window then shows the address of the server.

ILLUSTRATION

```
[1/4] Installing packages...
[2/4] Creating the start-up task...
[3/4] Opening port 8000 in Windows Firewall...
[4/4] Scheduling the daily backup...

Done. The server is starting and will start by itself every time this PC is switched on.

Open one of these addresses in a browser (this PC, or any PC on the school network):
    http://192.168.1.10:8000
    (on this PC you can also use http://localhost:8000)

First sign-in: admin / admin  - change the password straight away under Settings.
Data and backups are kept in: C:\IFPClassroom\data
```

What the installer sets up

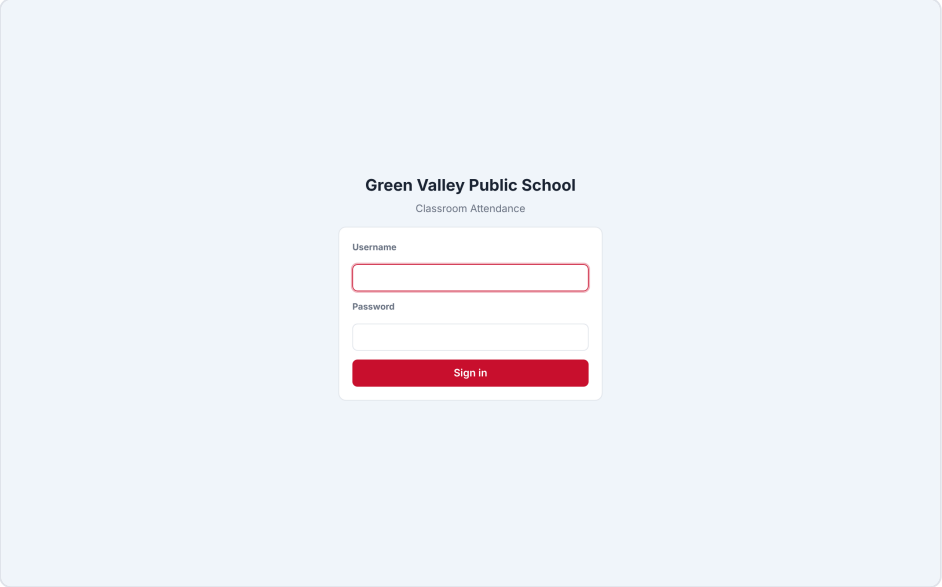
Item	Details
Start-up task *IFP Classroom Server*	Starts the server whenever Windows starts, even before anyone signs in. It restarts the server automatically if it ever stops.
Firewall rule *IFP Classroom Server*	Allows incoming TCP port 8000 on private and domain networks.
Task *IFP Classroom Backup*	Copies the database to `data\backups` every day at 17:30 and keeps the latest 30 copies.
`data\server.log`	Server messages, useful when something goes wrong.

If panels cannot connect later
Make sure the PC's network profile is **Private**, not Public (Windows Settings → Network → your connection). If another firewall or antivirus product is installed, allow port 8000 there too.

Running the installer again is safe.
It updates the program and keeps all data. Use it after copying in a newer version of the software.

4. First sign-in and the main screen

On the server PC open `http://localhost:8000`. On any other PC, open `http://<PC-IP>:8000`, e.g. `http://192.168.1.10:8000`. Sign in with **admin** / **admin**.



Sign-in page

> **Change the default password straight away**
> Go to **Settings** → *Change your password*. Also set your school's name under *School*; it appears on every page and in the Excel downloads.

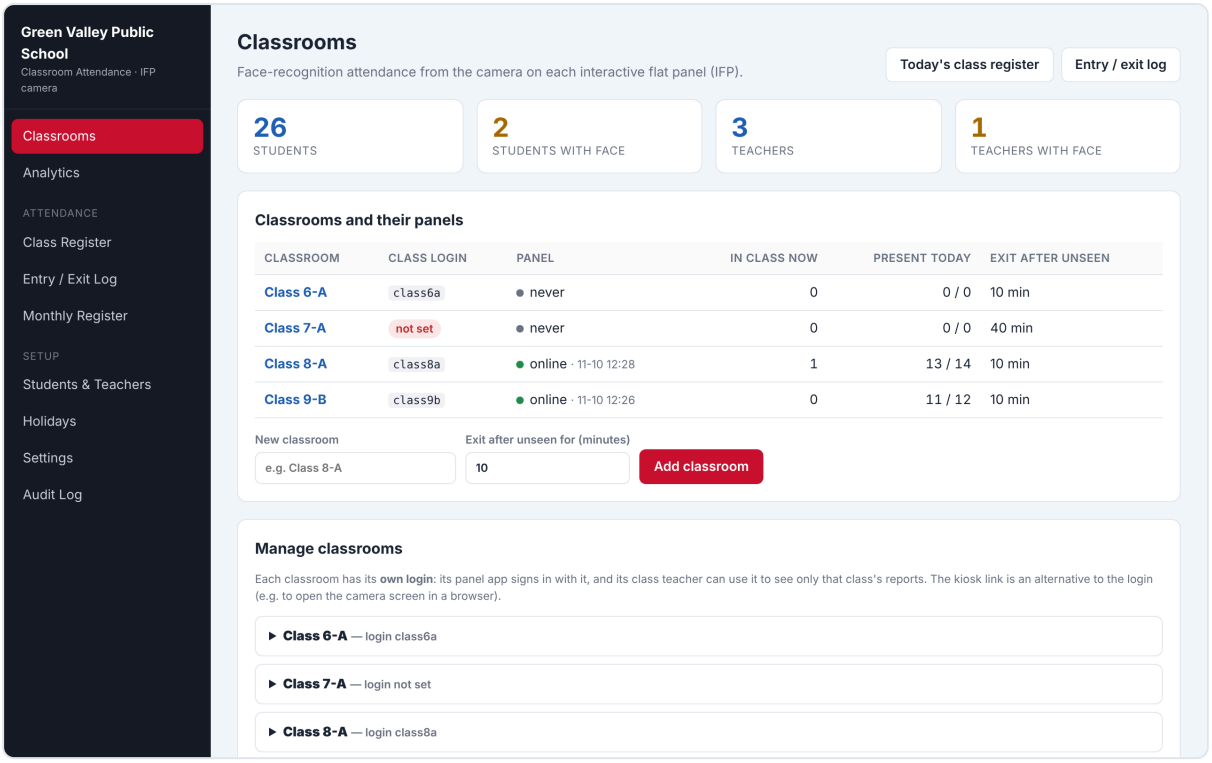
The menu

Menu item	What it is for	Chapter
Classrooms	Classrooms, their logins, panel status, periods and recognition settings	5, 6
Analytics	Attendance trends and summaries for any period	14
Class Register	Daily register for one classroom	11
Entry / Exit Log	Every entry and exit, with times	12
Monthly Register	Month grid of P / A / H / WO with totals and %	13
Students & Teachers	People, CSV import/export, face photos	7, 8
Holidays	School holidays	6
Settings	School name, your password, users, backup	15, 16
Audit Log	Who changed what, and when (admin only)	16

The pages also work on phones and tablets: tap **☰ Menu** to open the menu.

5. Classrooms and classroom logins

Open **Classrooms**. The top of the page shows the number of students and teachers and how many of them have face photos. The table below lists every classroom.



Classrooms page: summary and the classrooms table

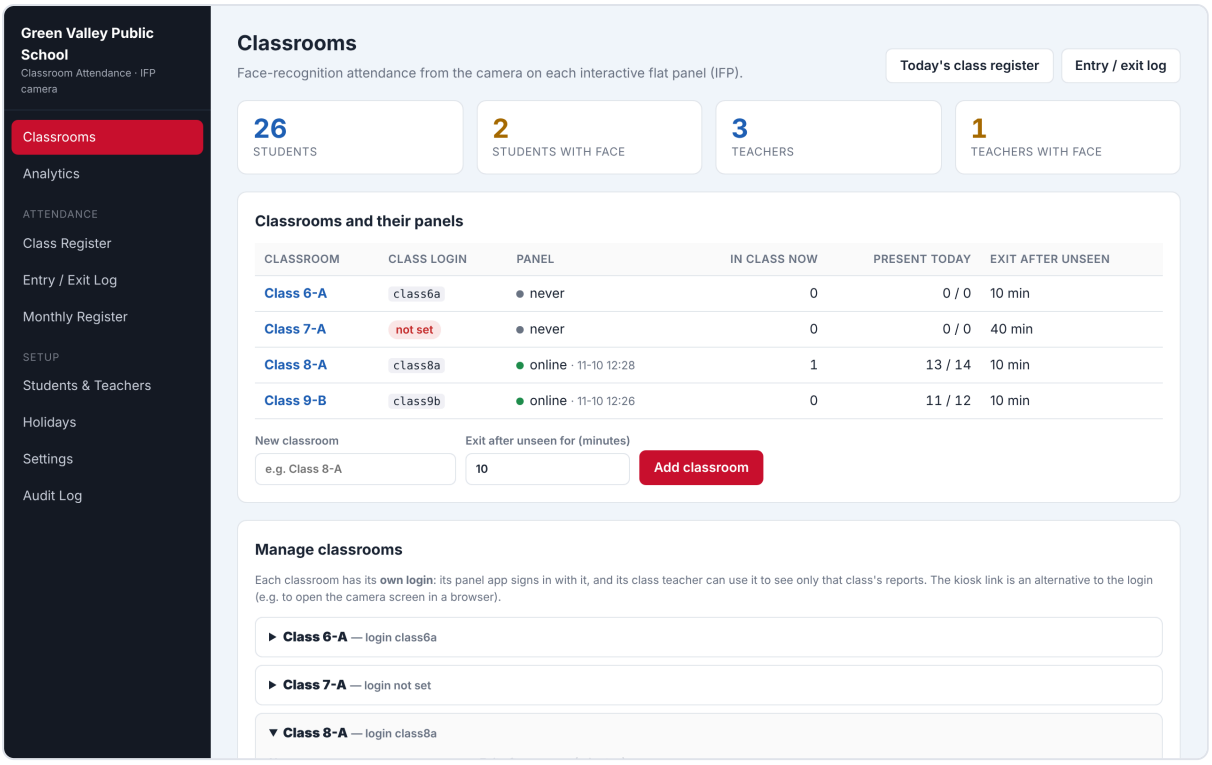
Column	Meaning
Classroom	Click the name to open today's register for that room.
Class login	The room's username. *not set* (red) means the panel cannot sign in yet.
Panel	**online**: the panel checked in during the last 10 minutes. **offline**: it has been silent longer. **never**: no panel has signed in yet. The time of the last check-in is shown.
In class now	People currently in the room.
Present today	Students of this home classroom seen today / total students of this room.
Exit after unseen	How long someone must be out of view before an exit is recorded.

Add a classroom

Below the table, type a name (e.g. *Class 8-A*), set *Exit after unseen for* (normally 10 minutes) and click **Add classroom**.

Give each classroom its own login

Under **Manage classrooms**, click a classroom to open it:



Manage classrooms: name and exit time, the classroom login, and the kiosk link

1. Type a **username** without spaces, e.g. `class8a`, and a **password** of at least 6 characters.
2. Click **Create login**. Note the username and password for the panel and the class teacher.
3. To change it later, type the new username and/or a new password and click **Update login**. Leaving the password empty keeps the old one.

Other actions for a classroom

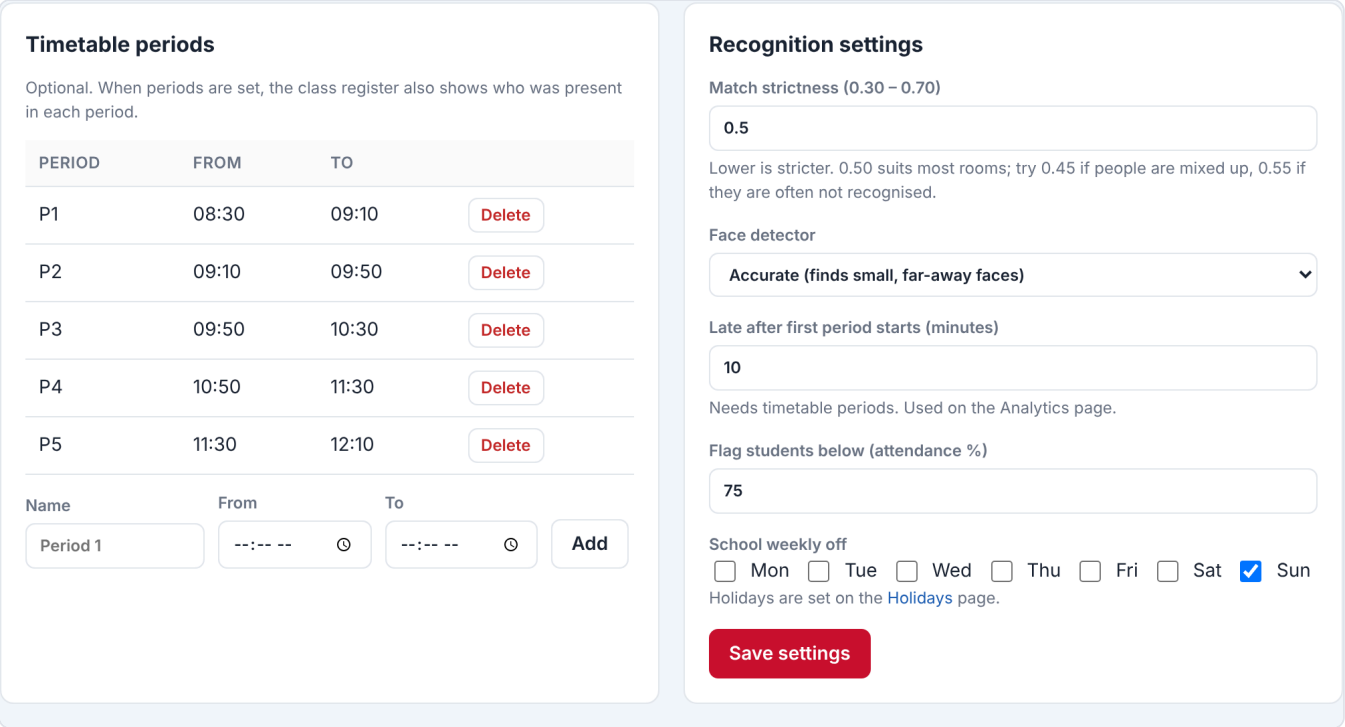
Action	Effect
Name / Exit after unseen / Status → **Save**	Rename the room, change its exit time, or *disable* it. A disabled room's panel stops recording.
Open kiosk	Opens the camera screen in a browser, for testing on a PC with a webcam.
New link (signs the panel out)	Replaces the room's secret kiosk link. The panel stops working until it signs in again. Use it if a login may have leaked, together with a new password.
Delete classroom	Deletes the room, its login and **all its entry/exit records**. It cannot be undone.

Choosing the exit time

10 minutes suits panels that watch the class all the time. If the panel only takes a roll call at the start of each period (see chapter 10), set it to the length of a period, e.g. 40 minutes. Otherwise students look as if they left mid-lesson.

6. Periods, holidays and recognition settings

These are at the bottom of the **Classrooms** page; holidays have their own page.



Timetable periods and recognition settings

Timetable periods (optional)

Add each period with a name, start and end time, e.g. *P1 08:30–09:10*. With periods set:

- the Class Register shows **P/A** per period (present = seen during that period);
- Analytics counts **late arrivals**: first seen after the first period's start plus the late grace.

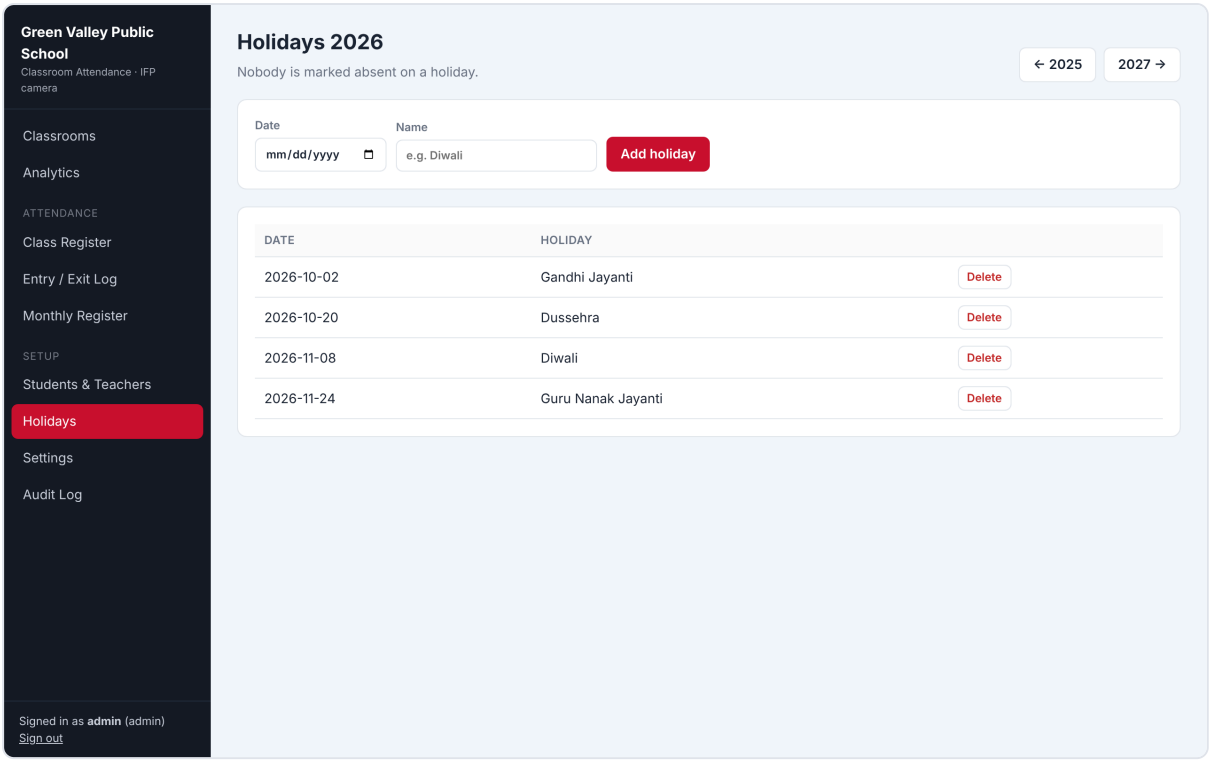
Recognition settings

Setting	Meaning	Default
Match strictness	How similar a face must be to an enrolled photo, from 0.30 to 0.70. **Lower is stricter.** If people get mixed up, lower it (e.g. 0.45). If people are often not recognised, raise it (e.g. 0.55).	0.50
Face detector	**Accurate** finds small and far-away faces. **Fast** suits slower panels. A panel switches itself to Fast if Accurate is too slow on it.	Accurate
Late after first period starts	Minutes of grace before an arrival counts as late (needs periods).	10
Flag students below	Attendance % under which a student is flagged on the Analytics page.	75%
School weekly off	Days with no school. Nobody is marked absent on them.	Sunday

Panels pick up changed settings within 5 minutes.

Holidays

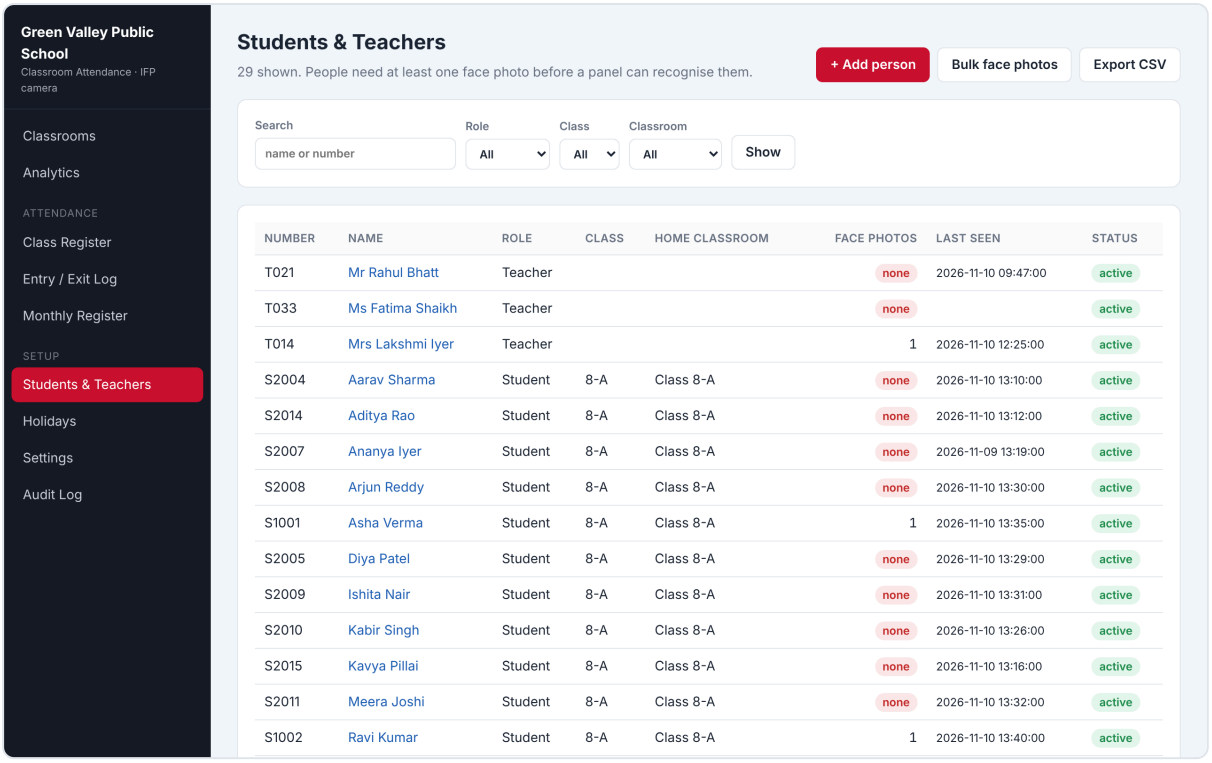
Open **Holidays**, choose a date and name and click **Add holiday**. Use the year buttons to move between years. Holidays show as **H** in the registers.



Holidays page

7. Students and teachers

Open **Students & Teachers**. You can search by name or number and filter by role, class or home classroom. The *Face photos* column shows **none** for people the panels cannot recognise yet.

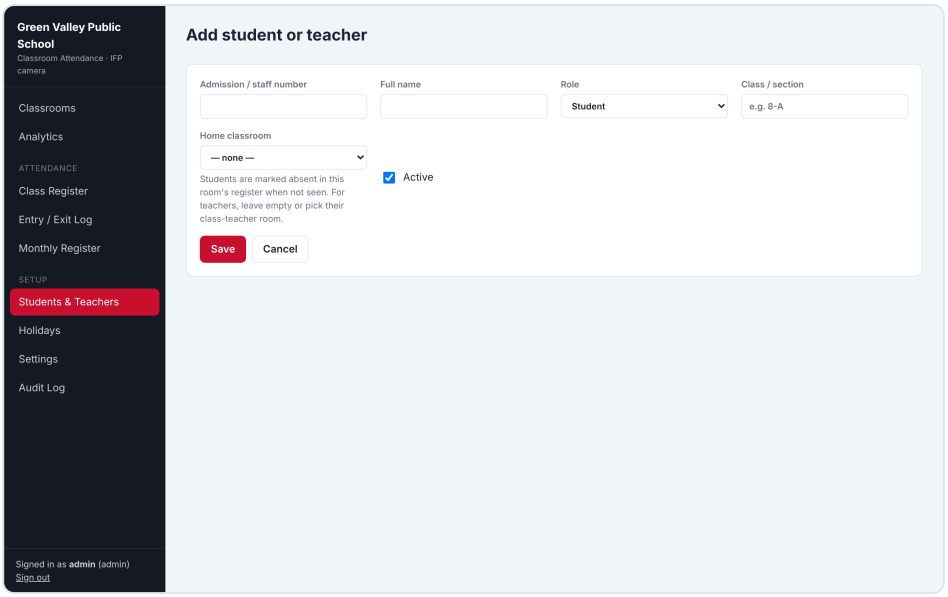


Students & Teachers list

Add one person

Click **+ Add person** and fill in the form:

Field	Notes
Admission / staff number	Unique, e.g. `S1023` or `T014`. Bulk photo files are named after it.
Full name	Shown on the panel and in reports.
Role	Student or Teacher.
Class / section	e.g. `8-A`. Used for filters, the monthly register and class-wise analytics.
Home classroom	The room whose register lists this student, and which marks them absent when they are not seen. For teachers, leave it empty or choose their class-teacher room.
Active	Untick for people who have left. They stop being recognised and listed, but their past records stay. To remove everything, use **Delete** on their page.



Add / edit person

Import many people from a CSV file

At the bottom of the list, choose a CSV file and click **Import**. Easiest: click **Export CSV** first to get a template, fill it in Excel, and save it as *CSV UTF-8*.

Number	Name	Role	Class	Classroom	Active
S1023	Asha Verma	student	8-A	Class 8-A	1
T014	Mrs Lakshmi Iyer	teacher			1

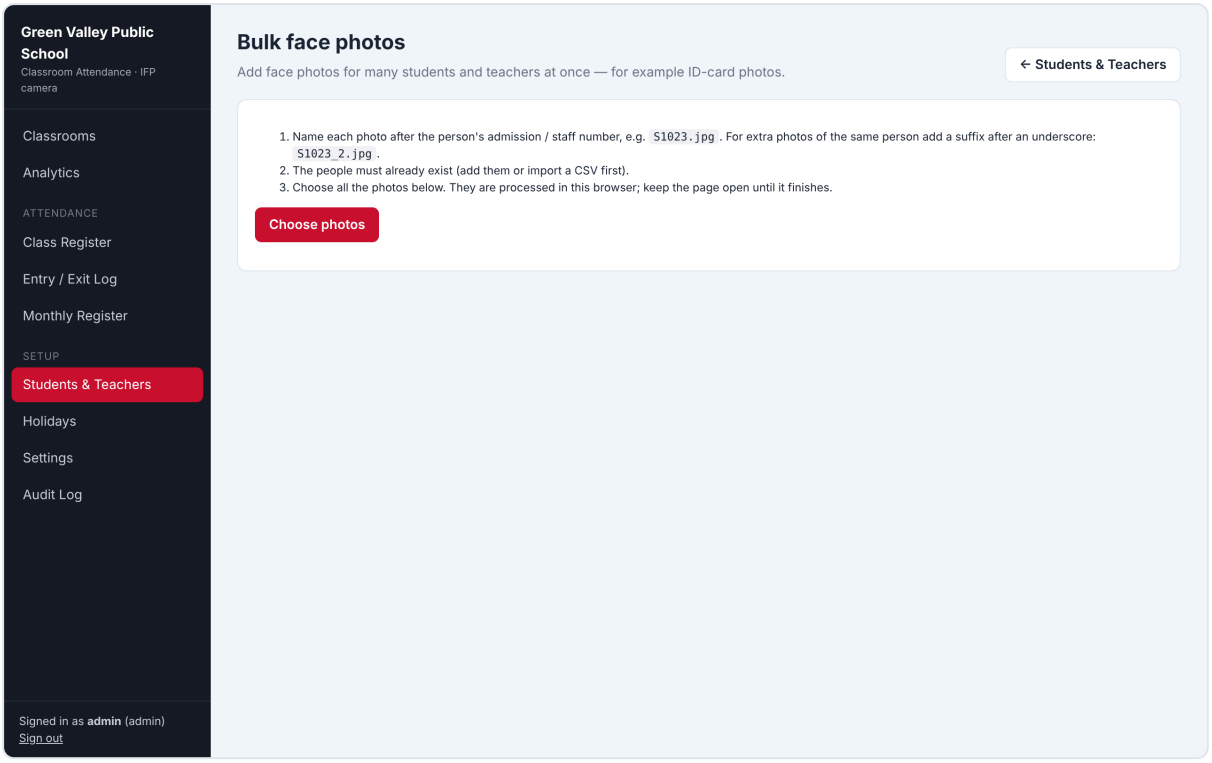
- *Classroom* must match a classroom name exactly; add classrooms first.
- Existing numbers are **updated**, new numbers are added. Problems are listed after the import.

8. Face photos (enrolment)

A panel can only recognise people who have at least one face photo. The best results come from **2–5 photos** per person: front-facing, good light, no sunglasses or masks. Photos are converted to face data **in your browser**. The server keeps only the face data (128 numbers) and a small thumbnail, never the original photo.

Option A: bulk upload (fastest for a whole school)

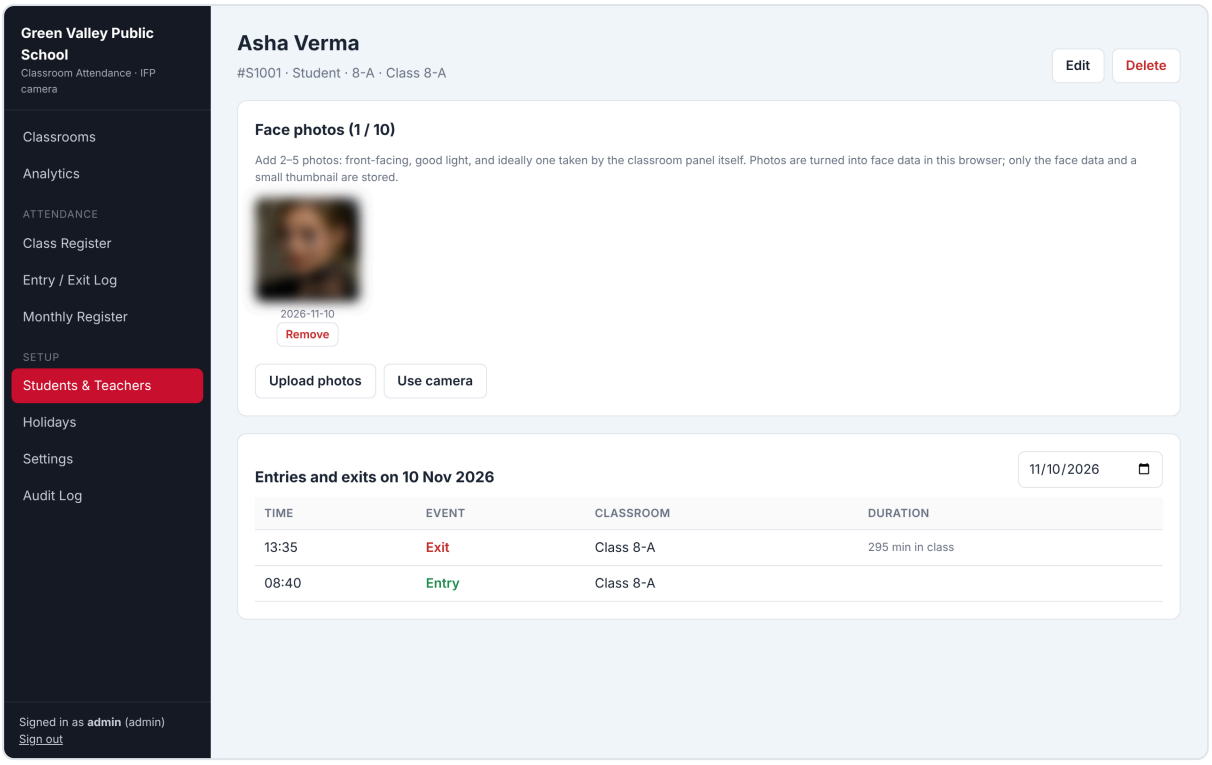
1. Name each photo file after the person's number, e.g. `S1023.jpg`. For extra photos of the same person add `_2`, `_3` …, e.g. `S1023_2.jpg`.
2. Make sure the people already exist (chapter 7).
3. Open **Students & Teachers** → **Bulk face photos** → **Choose photos** and select all the files. Keep the page open until it says *Done*.



Bulk face photos: each file is listed as added (✓) or with the reason it failed (×)

Option B: one person, from files or a camera

Open the person's page. Use **Upload photos** to choose image files, or **Use camera** → **Capture face** to take photos with a webcam. Turn the head slightly between captures. Photos can be removed with **Remove**. Each person can have up to 10.



Person page: face photos (blurred here), enrolment buttons and the day's entries and exits

> **Tip: one photo from the panel itself**
> Recognition is best when one photo comes from the same camera. Sign in to the server in the panel's browser, open the person's page and use **Use camera**.

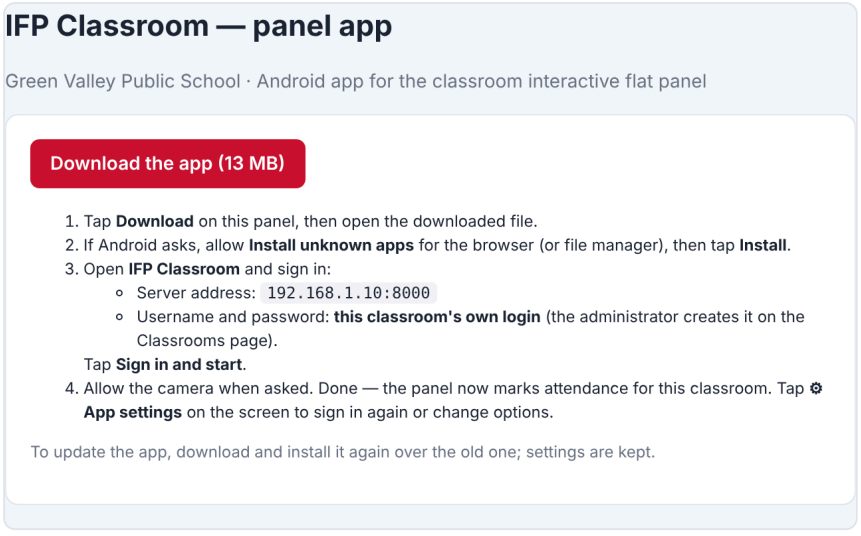
Why a photo can be rejected

Message	What to do
no face found	Use a clearer, front-facing photo with the face fully visible.
more than one face in the photo	Crop the photo to one person.
the face is too small	Use a closer photo; the face should be at least about 60 pixels wide.
No student or teacher with number ...	The file name does not match any number; check it, or add the person first.
already has 10 face photos	Remove an old photo first.

9. Install the app on each panel

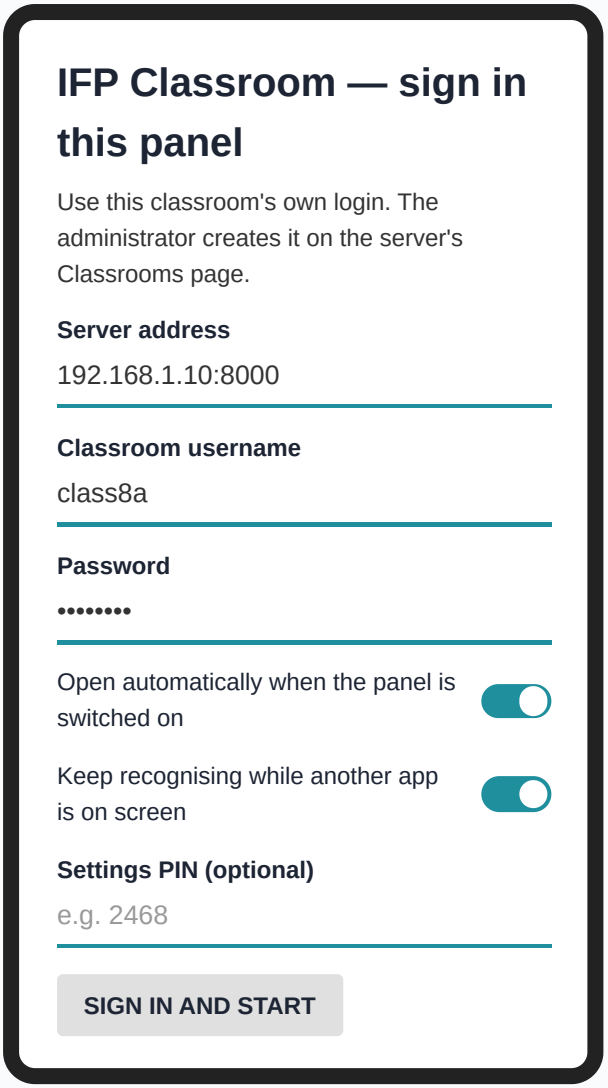
Do this once on every classroom panel. It takes about 5 minutes per panel. Have that classroom's login ready (chapter 5).

1. On the panel, open its web browser and go to `http://<PC-IP>:8000/app`, e.g. `http://192.168.1.10:8000/app`. Tap **Download the app**.
2. Open the downloaded file. If Android asks, allow **Install unknown apps** for the browser or file manager, then tap **Install**.
3. Open **IFP Classroom** and fill in the sign-in screen (below), then tap **Sign in and start**.
4. Tap **Allow** when Android asks for the camera. The camera screen appears and attendance starts.



The download page as seen on a panel

ILLUSTRATION OF THE APP'S SIGN-IN SCREEN



Field / option	Notes
Server address	The server PC's address. `192.168.1.10` alone also works; port 8000 is assumed.
Classroom username / password	This room's login. An admin or viewer login is refused, so a panel is always linked to one classroom. The password is **not stored** on the panel.
Open automatically...	Opens the camera screen when the panel is switched on (see below).
Keep recognising...	Tries to keep the camera working while a teacher uses another app (see chapter 10).
Settings PIN	Asked for before anyone can change these settings from the camera screen. Recommended.

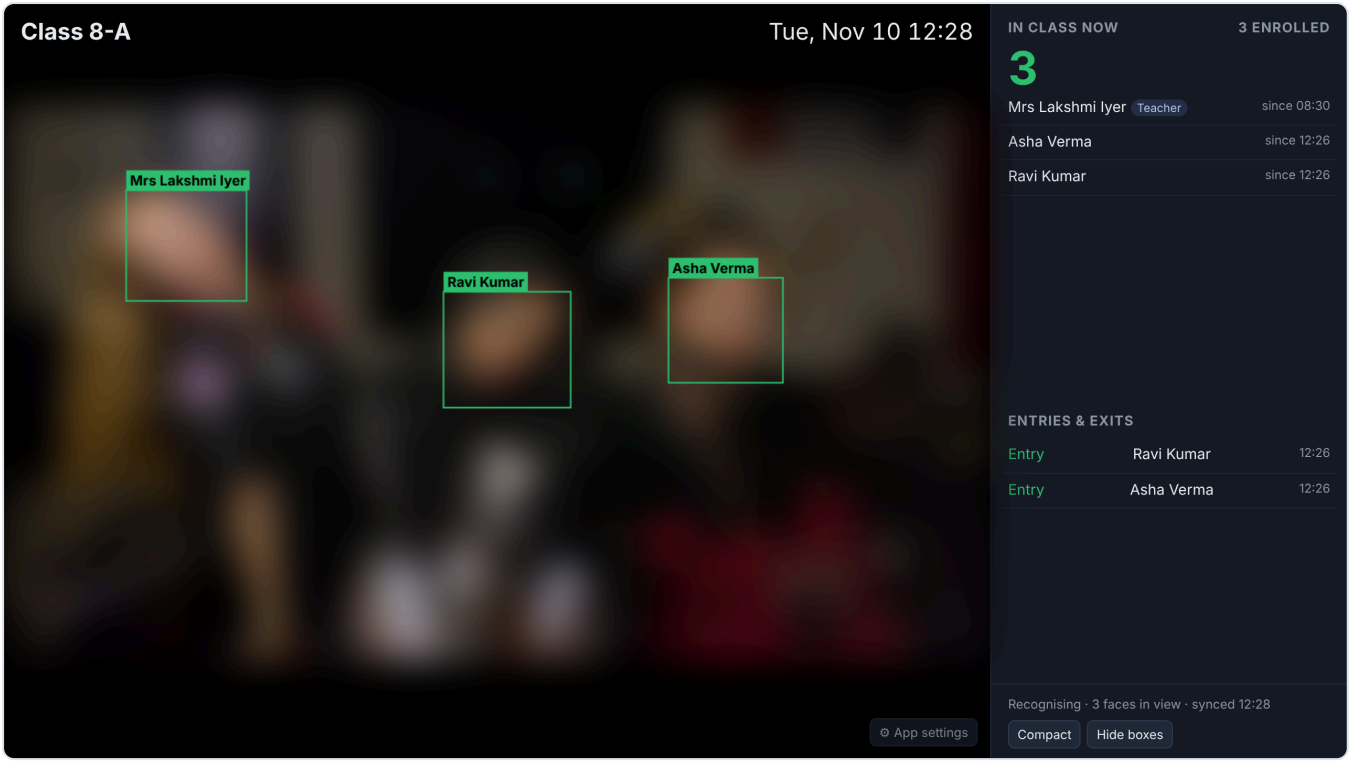
Automatic start when the panel is switched on

- On **Android 10 and later**, tap **Allow "Display over other apps"** on the sign-in screen and switch it on for IFP Classroom. Android needs this before an app can open itself at power-on.
- Many panels also have their own *Boot app* or *Start-up app* setting; choose IFP Classroom there if available.

Changing the classroom, signing in again, updating

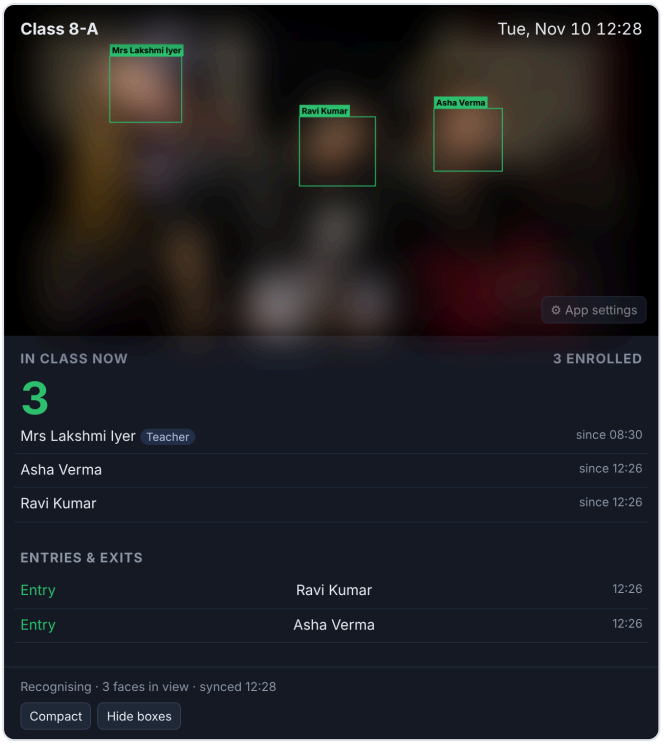
- On the camera screen, tap **⚙ App settings** and enter the PIN if one is set. Sign in with another classroom's login, or tap **Save options only** to change only the options.
- If the screen says the panel is *no longer signed in*, the room's link was replaced or the room was removed. Sign in again.
- **To update the app**, download and install it again from `/app` over the old one. The panel stays signed in.

10. The camera screen on the panel



Camera screen of the panel app (camera image blurred for this guide)

Part	What it shows / does
Top bar	Classroom name, date and time.
Boxes on faces	**Green with a name**: recognised. **Yellow "Unknown"**: a face with no matching enrolment. A person is counted after being recognised twice within 10 seconds, which avoids one-frame mistakes.
In class now	Count and names of people in the room, teachers first, each with the time since they entered.
Entries & exits	The latest entries and exits seen by this panel.
Status line	Recognising / faces in view / last sync time / records waiting to be sent. Problems such as the server being unreachable appear here in red.
Compact	A narrower layout for a small or side window.
Hide boxes	Hides the boxes and names on the camera image; recognition continues.
Camera list	Appears when the panel has more than one camera. The choice is remembered.
⚙ App settings	Opens the sign-in / options screen (PIN protected).



Compact layout, for split screen or a small window

During lessons

Android limits camera use by an app that is not on screen. With *Keep recognising while another app is on screen* switched on, the app asks Android to keep the camera running. A "Classroom attendance is running" notification shows while it does. How much Android allows depends on the panel's Android version and maker, so **test it on your panel model**:

1. Open the app and let it recognise you.
2. Switch to the whiteboard app for 5 minutes and walk in front of the camera.
3. On the server, check the **Entry / Exit Log**.

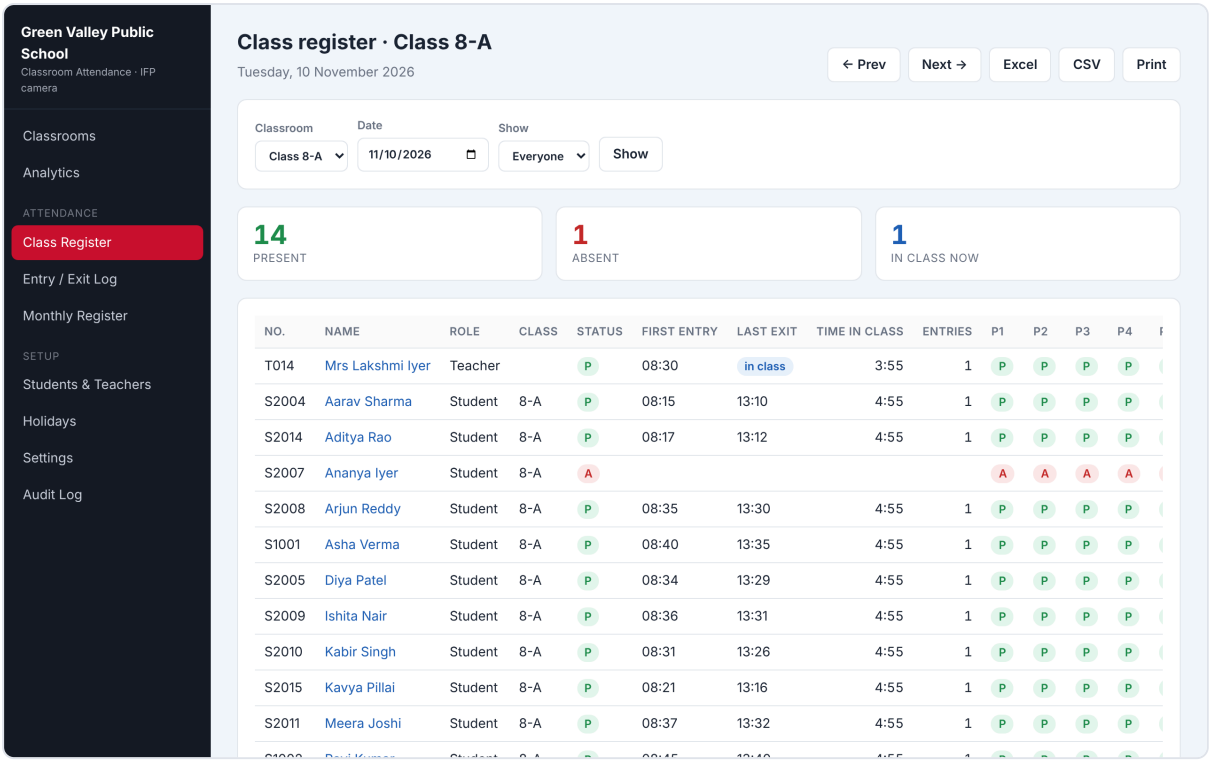
If that doesn't work on your panel model:

- **Split screen / floating window**: keep the app (Compact) beside the lesson.
- **Roll call**: keep the app on screen for the first 2–3 minutes of each period. Set the room's exit time to the period length.

Network or server down?
The panel keeps recognising and stores its records, sending them when the connection returns; the status line shows how many are waiting. Times are corrected by the server, so a wrong panel clock does not matter.

11. Class register

Open **Class Register**. Choose the classroom, the date and whether to show everyone, only students or only teachers. Use **← Prev** / **Next →** to move day by day.



Class register for one classroom and day

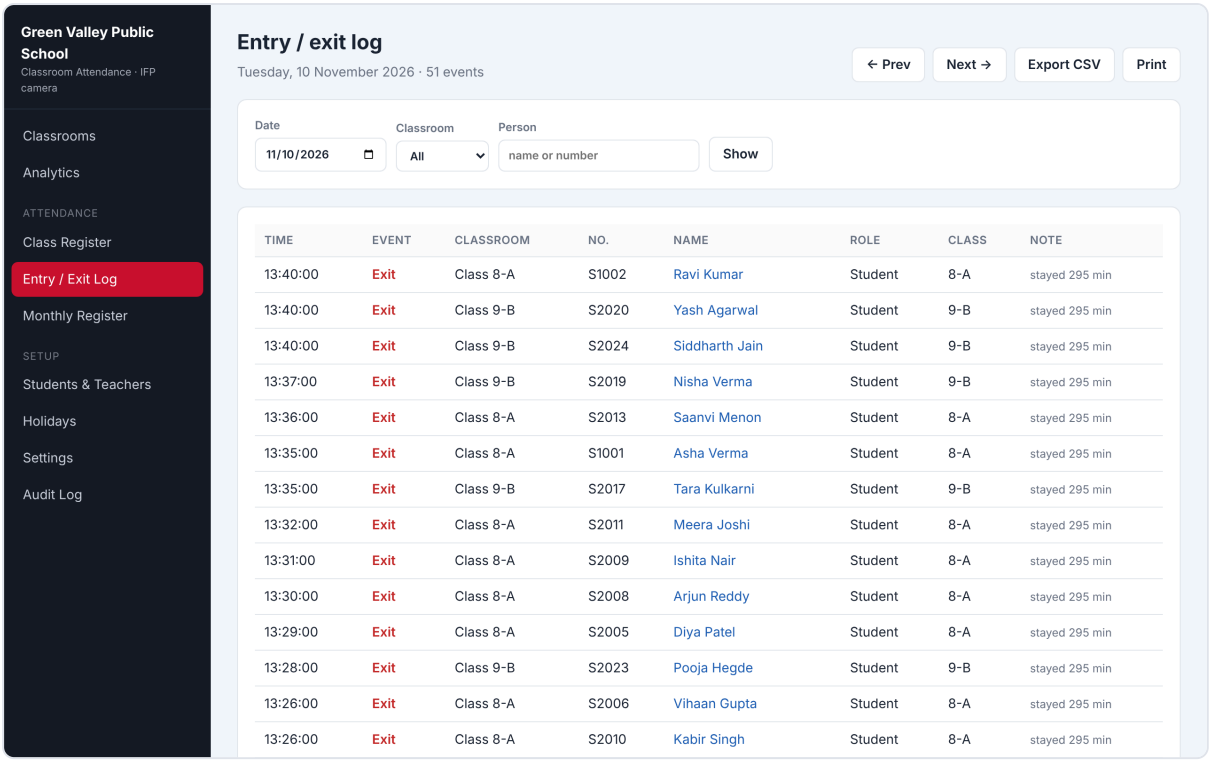
Column	Meaning
Status	**P** present (seen in this room that day) · **A** absent · **H** holiday · **WO** weekly off · – future day.
First entry	First time seen in this room that day.
Last exit	Last time seen, or **in class** if they are still in the room.
Time in class	Total time between each entry and its exit (hours:minutes).
Entries	How many times they came in. More than 1 means they left and returned.
P1, P2…	Per-period attendance (only when periods are set): P if seen during the period.

The register lists the room's **home students** (absent ones included) plus **anyone else seen in the room**, such as subject teachers or students from another section.

Downloads: **Excel** (formatted sheet), **CSV** and **Print** (print-friendly layout, or save as PDF from the print dialog).

12. Entry / exit log and person page

Open **Entry / Exit Log** to see every entry and exit for a day, newest first. Filter by classroom or by a person's name or number. **Export CSV** downloads the list in time order.



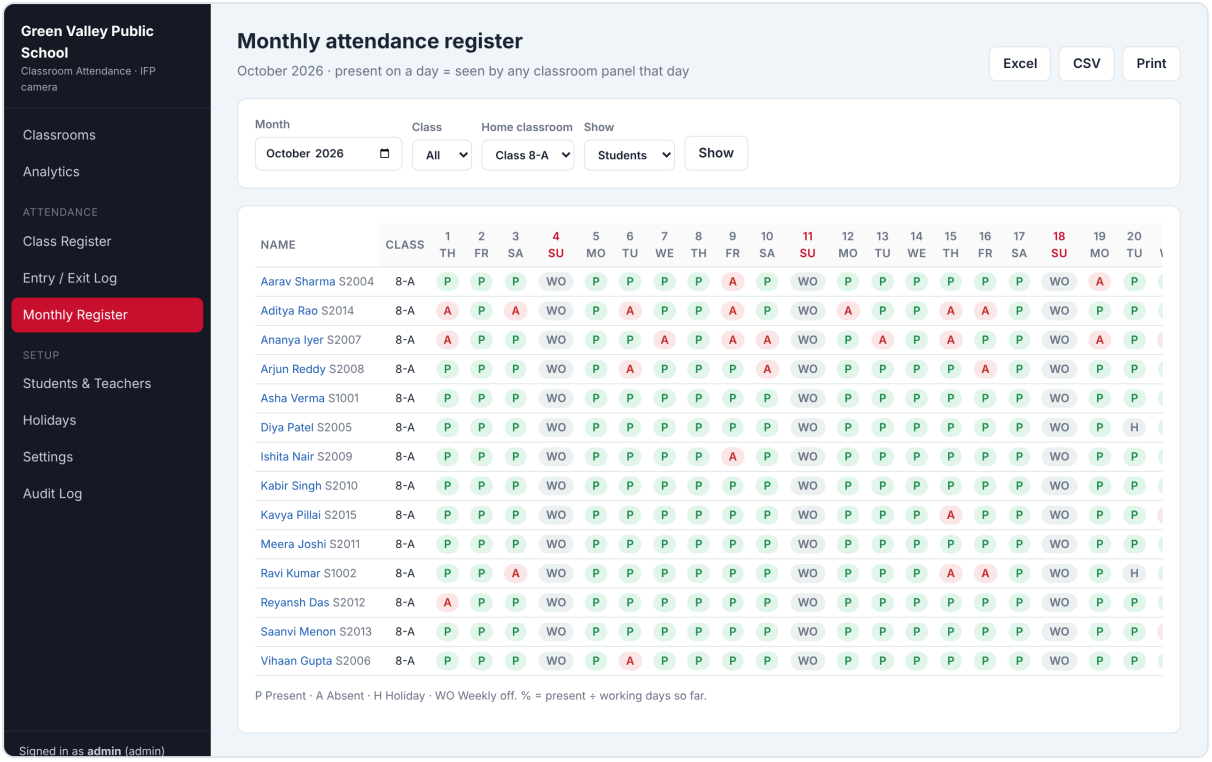
Entry / exit log

- Each **Exit** row shows how long the person stayed. *in class now* marks someone who has not left yet.
- Click a name to open the **person page**. It shows their face photos and their entries and exits on any date (choose the date at the top right of that section).

> **Why is an exit time shown later than expected?**
> An exit is only recorded once the person has been out of view for the room's exit time. The exit time itself is the *last moment they were seen*, not the moment the exit was recorded.

13. Monthly register

Open **Monthly Register** and choose the month. You can filter by class or home classroom, and choose students, teachers or everyone. A day counts as **P** when the person was seen by *any* classroom panel that day.

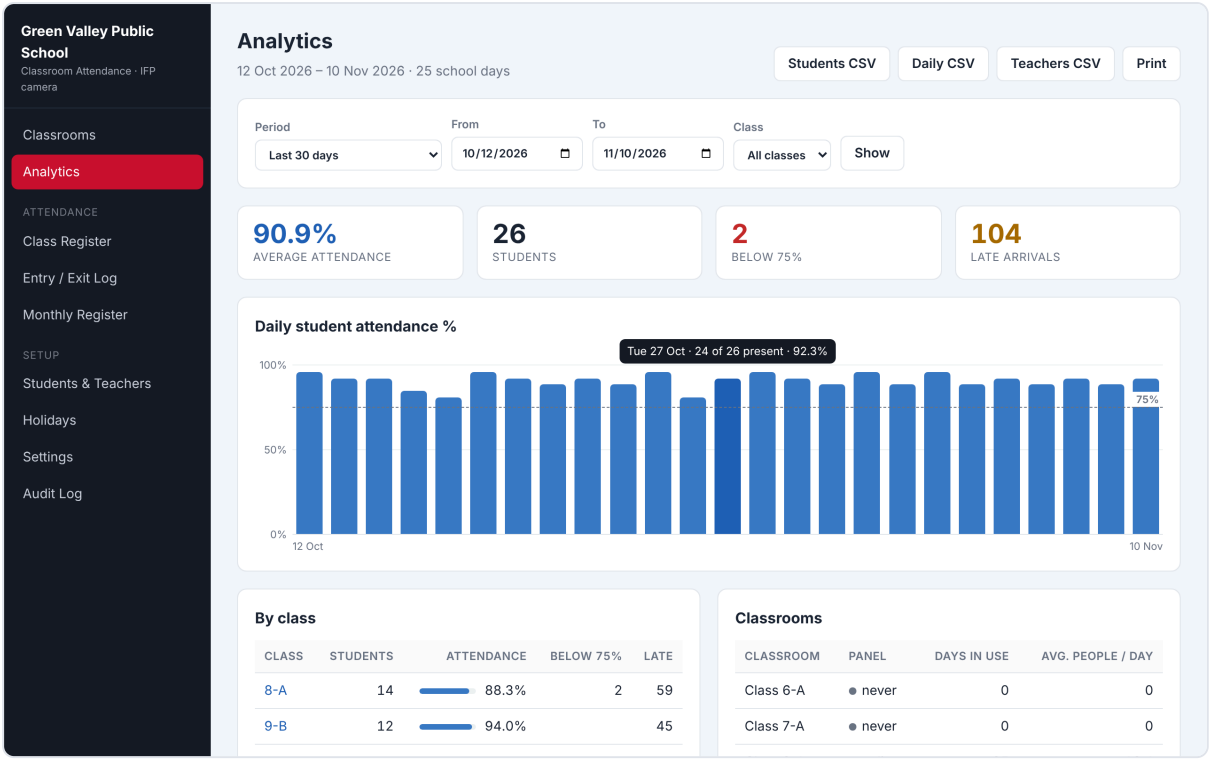


Monthly register (October 2026, Class 8-A)

- **P** present · **A** absent · **H** holiday · **WO** weekly off · blank: future day.
- **%** = days present ÷ working days so far (holidays and weekly offs are not counted).
- **Excel** gives a colour-coded sheet ready to print or file. **CSV** gives the same data for other software.

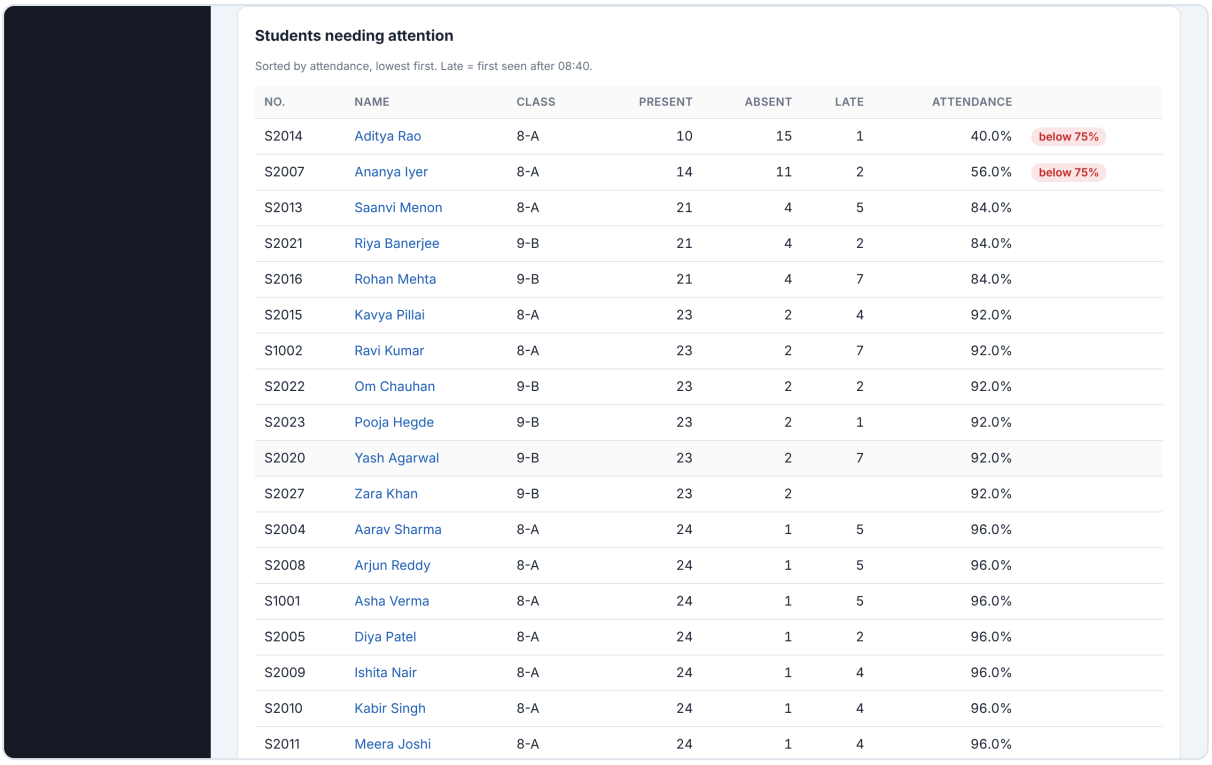
14. Analytics

Open **Analytics**. Choose a **period** (this week, this month, the last 30 days, the academic year from April, or custom dates) and optionally a **class**. Only school days are counted: holidays and weekly offs are left out.



Analytics: headline numbers and the daily attendance chart (hover a bar for details)

Section	What it shows
Average attendance	Student-days present ÷ possible student-days in the period.
Below 75%	Number of students under the low-attendance threshold (chapter 6).
Late arrivals	Student-days first seen after the first period start plus the grace (needs periods).
Daily student attendance %	One bar per school day; the dashed line marks the threshold. Point at a bar to see the date and how many were present.
By class	Attendance %, number below threshold, and late arrivals per class. Click a class to analyse only that class.
Classrooms	Panel status, the days each room was in use, and the average number of people per day.
Students needing attention	Students sorted by attendance, lowest first, with absences and late arrivals; the lowest 50 are shown.
Teachers — time in classrooms	Days seen, total and average daily time each teacher spent in classrooms.



Students needing attention

Sorted by attendance, lowest first. Late = first seen after 08:40.

NO.	NAME	CLASS	PRESENT	ABSENT	LATE	ATTENDANCE	
S2014	Aditya Rao	8-A	10	15	1	40.0%	below 75%
S2007	Ananya Iyer	8-A	14	11	2	56.0%	below 75%
S2013	Saanvi Menon	8-A	21	4	5	84.0%	
S2021	Riya Banerjee	9-B	21	4	2	84.0%	
S2016	Rohan Mehta	9-B	21	4	7	84.0%	
S2015	Kavya Pillai	8-A	23	2	4	92.0%	
S1002	Ravi Kumar	8-A	23	2	7	92.0%	
S2022	Om Chauhan	9-B	23	2	2	92.0%	
S2023	Pooja Hegde	9-B	23	2	1	92.0%	
S2020	Yash Agarwal	9-B	23	2	7	92.0%	
S2027	Zara Khan	9-B	23	2		92.0%	
S2004	Aarav Sharma	8-A	24	1	5	96.0%	
S2008	Arjun Reddy	8-A	24	1	5	96.0%	
S1001	Asha Verma	8-A	24	1	5	96.0%	
S2005	Diya Patel	8-A	24	1	2	96.0%	
S2009	Ishita Nair	8-A	24	1	4	96.0%	
S2010	Kabir Singh	8-A	24	1	4	96.0%	
S2011	Meera Joshi	8-A	24	1	4	96.0%	

Analytics: by class, classrooms, and students needing attention

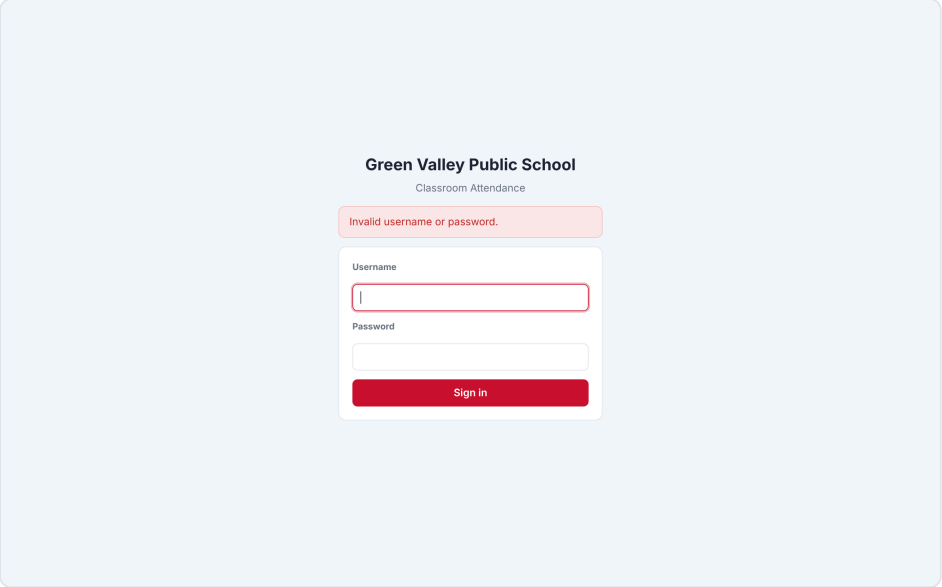
Downloads: **Students CSV** (every student's present/absent/late/%), **Daily CSV** (attendance % per day) and **Teachers CSV** (time in classrooms). Open them in Excel for your own charts and pivot tables.

15. Users, roles and the class teacher's view

Login type	Who	Sees	Can change	Created at
Admin	You (IT administrator)	Everything	Everything	Settings → Users
Viewer	Principal, office	Every class's reports and analytics	Only their own password	Settings → Users
Classroom	That room's panel app and its class teacher	Only that class: register, entry/exit log, monthly register, analytics, its own students' pages	Only its own password	Classrooms → Manage classrooms

What a class teacher sees

When the class teacher signs in with the classroom login (e.g. `class8a`) in a browser, the menu shows only *My classroom* pages, limited to that room. Pages of other classes and all setup pages are not available.

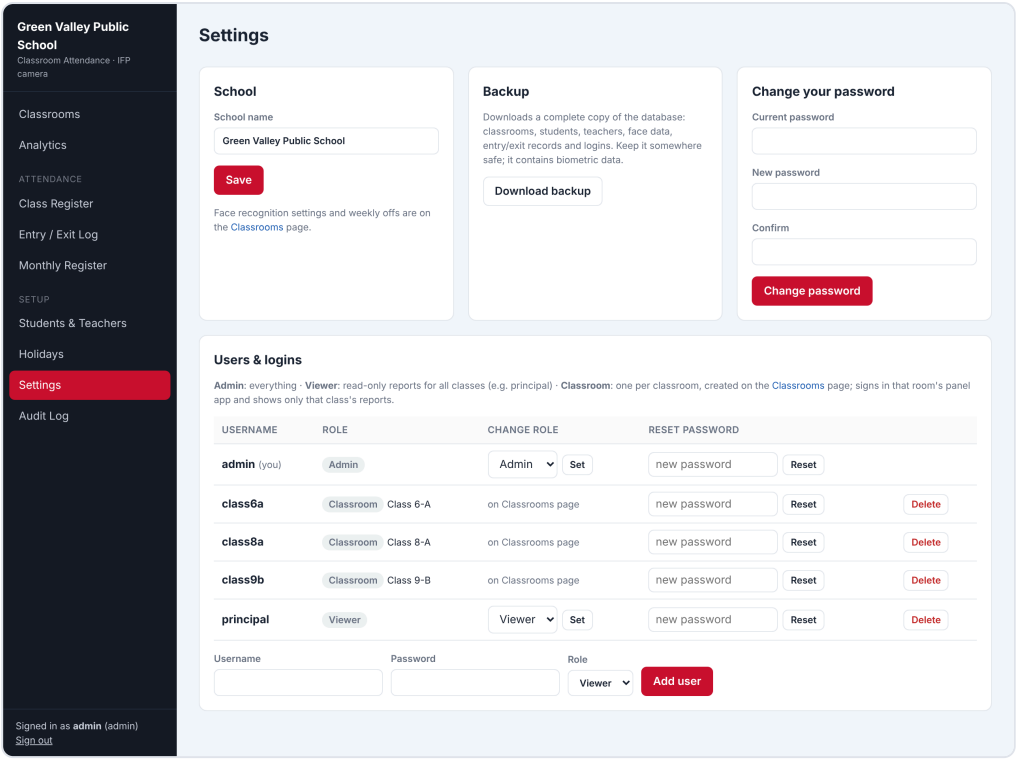


Signed in with a classroom login: only that class's pages

Managing users (Settings → Users & logins)

- **Add user**: username, password (6+ characters) and role (Admin or Viewer).
- **Change role** (Admin ↔ Viewer), **Reset password**, **Delete**. At least one admin must remain, and you cannot delete yourself.
- Classroom logins are listed with their room; change them on the Classrooms page.
- After 5 wrong passwords, sign-in is blocked for 15 minutes. Panel sign-in is blocked after 10 wrong attempts in 15 minutes.

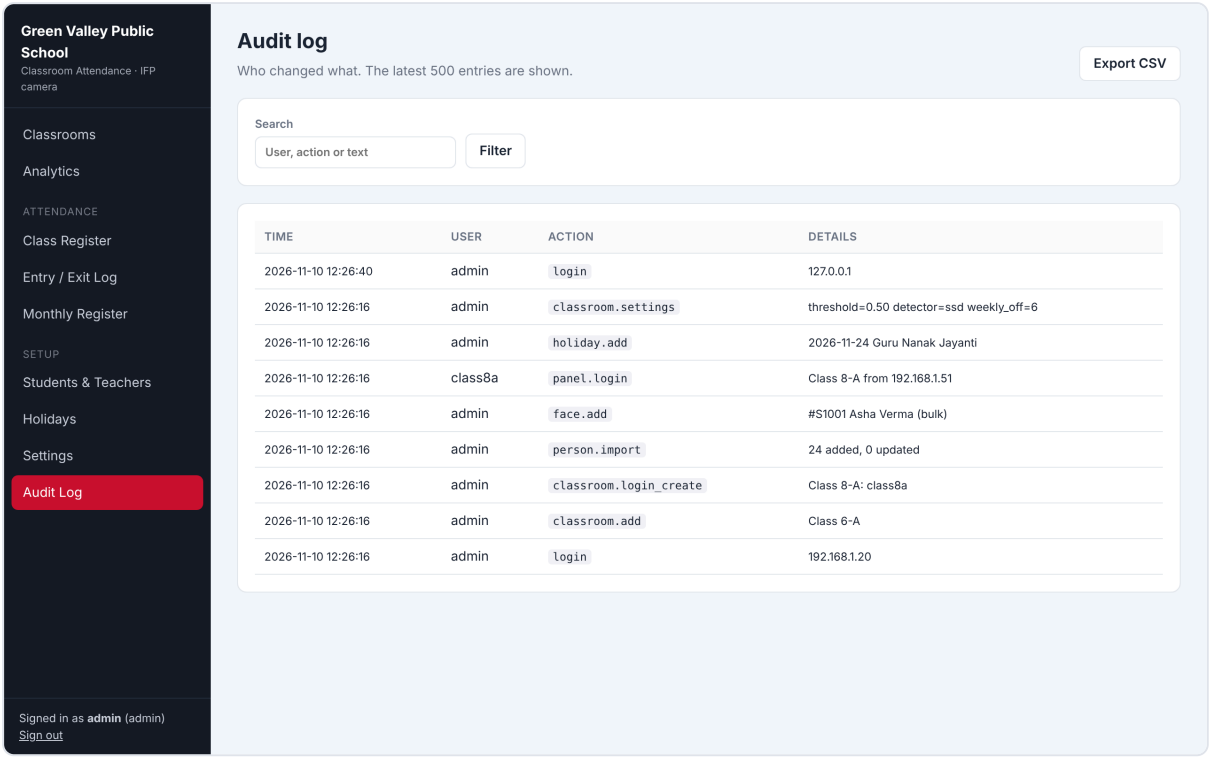
16. Settings, audit log, backup and updates



Settings: school name, backup, password and users

Audit log

Audit Log (admin only) records who did what and when: sign-ins, panel sign-ins, classrooms, logins, people, face photos, holidays and settings. Search it, or download it with **Export CSV**.



Audit log

Backup and restore

- **Automatic:** every day at 17:30 into `data\backups` (latest 30 kept), and always before an update changes the database.
- **Manual:** Settings → **Download backup**. Also copy the whole `data` folder to another disk or drive now and then.
- **Restore:** run `windows\uninstall_windows_server.bat` as administrator to stop the server. Copy the backup file over `data\classroom.db` (rename it to that name), then run the installer again.

> **Backups contain biometric data.**
> Store them as carefully as the server itself.

Updating the software

1. Copy the new version of the `ifp-classroom` folder over the old one. Do **not** delete the `data` folder.
2. Run `windows\install_windows_server.bat` as administrator again. Database changes are applied automatically after a safety backup.
3. If the panel app changed, reinstall it on each panel from `/app`. It installs over the old one.

Moving to another PC or uninstalling

Run `windows\uninstall_windows_server.bat` as administrator. It removes the start-up task, the backup task and the firewall rule; the `data` folder is kept. To move, copy the whole folder, including `data`, to the new PC and run the installer there. If the IP address changes, sign each panel in again with the new address.

Forgotten admin password

On the server PC, open a command prompt in the `ifp-classroom` folder and run: `set CLASSROOM_DATA_DIR=%cd%\data` then `.venv\Scripts\python -m flask --app classroom_app reset-admin`. The admin password becomes **admin**; change it straight away.

17. Troubleshooting

Problem	What to check / do
Installer says "Python 3.10 or newer was not found"	Install Python from python.org with **Add Python to PATH** ticked. If it is installed, switch off the Store shortcuts: Windows Settings → Apps → Advanced app settings → App execution aliases → turn off *python.exe* and *python3.exe*. Run the installer again.
Browser cannot open `http://<PC-IP>:8000`	Is the server PC on? Does `http://localhost:8000` work on the PC itself? If yes, the firewall or network profile is blocking others (chapter 3). If no, look at `data\server.log` and run the installer again.
Panel: "Cannot reach the server"	Is the address right (with `:8000`)? Is the panel on the school network rather than a guest Wi-Fi? Can the panel's browser open `http://<PC-IP>:8000/app`?
Panel: "Wrong username or password"	Check the classroom login on the Classrooms page. Set a new password with **Update login** if needed.
Panel: "Use the classroom's own login"	An admin or viewer login was used; use the classroom's login.
Panel: "no longer signed in"	The room's link was replaced (**New link**) or the room was deleted or disabled. Sign in again.
Panel shows *offline* on Classrooms	The app is closed, the panel is off, or the network is down. Open the app; the status line shows the reason.
Camera does not start	Allow the camera for IFP Classroom (Android Settings → Apps → IFP Classroom → Permissions). Close other apps using the camera, e.g. video calls. If the panel has two cameras, choose the right one in the camera list.
"No students or teachers have face photos yet"	Add face photos (chapter 8). The panel picks them up within 5 minutes.
Someone is not recognised	Add more photos, including one from the panel. Check lighting and distance (faces up to about 5–6 m). Raise *Match strictness* slightly (0.55).
Two people are confused	Lower *Match strictness* (0.45) and add more photos of both.
Students marked as leaving mid-lesson	The panel could not see them, e.g. while another app was full screen. Increase the room's exit time, or use split screen (chapter 10).
App does not open at power-on	Allow "Display over other apps" (chapter 9) and/or set the panel's own start-up app setting.
Recognition is slow on a panel	Set *Face detector* to Fast (chapter 6). The panel also does this by itself when it detects it is too slow.
Wrong attendance on a holiday	Add the date on the Holidays page; reports update immediately.

18. Privacy, security and limits

Consent and transparency

- The system processes **biometric data of children**. Under India's Digital Personal Data Protection Act 2023, the school must obtain verifiable consent from a parent or guardian before enrolling a student. It must also tell staff and parents what is collected, why, and for how long.
- Do not enrol anyone without consent. Put up a notice in classrooms with panels.

What is stored, and where

- Only on the server PC: face data (numbers), a small face thumbnail, and the times each person was seen in each classroom. No video or full photos are stored.
- Delete students and staff who leave (their page → **Delete**); this removes their face data and records. Remember that older backups still contain them.

Security

- Change the default admin password; use strong passwords for classroom logins.
- A classroom login, and its kiosk link, gives access to that room's face data. If one may have leaked, set a new password and click **New link** for that room, then sign the panel in again.
- Use the PIN in the panel app so students cannot change its settings.
- Keep the server PC locked and updated; it holds all the data.

Limits

- **Not proof against spoofing:** a clear photo of a person held up to the camera can be recognised as that person. Treat the registers as attendance support, not a security control.
- Recognition is very good but not perfect. Faces far away, turned away, poorly lit or covered may be missed. Check absences before acting on them.
- Background recognition while other apps are open depends on the panel model (chapter 10).

19. Go-live checklist

✓	Task	Chapter
☐	Parent/guardian consent collected; notice displayed	18
☐	Server PC has a fixed IP, does not sleep, network profile is Private	2, 3
☐	Server installed; `http://<PC-IP>:8000` opens from another PC	3
☐	Admin password changed; school name set	4
☐	All classrooms added, each with its own login	5
☐	Weekly offs, holidays and timetable periods set	6
☐	Students and teachers imported with home classrooms	7
☐	Everyone has at least 2 face photos (no "none" on the people list)	8
☐	Pilot: one panel installed, signed in, recognising, and showing *online*	9, 10
☐	"During lessons" test done on the panel model; exit times chosen accordingly	10
☐	Registers checked against a manual roll call for a few days	11–13
☐	App installed on all panels with a PIN; auto-start checked after a power cycle	9
☐	Class teachers given their classroom login; principal given a viewer login	15
☐	Backup checked: a file appears in `data\backups`; a copy is kept off the PC	16

Appendix A. Reference

A.1 People CSV columns

Column	Required	Values
Number	yes	Admission or staff number; unique. Existing numbers are updated.
Name	yes	Full name.
Role	no	`student` (default) or `teacher`.
Class	no	Class/section, e.g. `8-A`.
Classroom	no	Home classroom; must match a classroom name.
Active	no	`1` (default) or `0` / `no` / `inactive`.

A.2 How attendance is calculated

- A panel records a person after recognising them twice within 10 seconds, then at most once every 30 seconds while they stay in view.
- Sightings of one person in one room are grouped into **visits**: a gap longer than the room's exit time starts a new visit. Visit start = entry, visit end = exit; time in class is the sum of the visits.
- **Class register:** present if seen in that room that day. **Period:** present if seen between the period's start and end.
- **Monthly register and Analytics:** present on a day if seen in any room that day. Working days exclude holidays, weekly offs and future days.
- **Late:** first seen that day after the first period's start plus the late grace.

A.3 Settings reference

Setting	Where	Range / default
Exit after unseen (per room)	Classrooms	1–240 minutes; 10
Match strictness	Classrooms	0.30–0.70; 0.50 (lower = stricter)
Face detector	Classrooms	Accurate / Fast; Accurate
Late grace	Classrooms	0–120 minutes; 10
Low-attendance threshold	Classrooms	1–100 %; 75
Weekly off	Classrooms	any days; Sunday
School name	Settings	text

A.4 Addresses

Address	Purpose
`http://<PC-IP>:8000`	Server web pages (sign in)
`http://<PC-IP>:8000/app`	Panel app download page (no sign-in needed)
`http://<PC-IP>:8000/kiosk/<code>`	A classroom's camera screen in a browser (kiosk link)